



**Programme: B E – Computer Science and Engineering (AI&ML) &
Computer Science and Engineering (Cyber Security)
Internal Assessment – II**

TERM : 3 rd October 2024 to 25 th Jan 2025	COURSE NAME: Computer Organization and Architecture
DATE : 22-01-2025 TIME: 11:00 to 12:00 NOON	COURSE CODE : CI34/CY34
MAX MARKS: 30	PORTIONS : L20 to L42



Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Explain the direct-mapped, set-associative, and fully associative cache mapping techniques. Assume a system with 128 KB of main memory and a cache of 32 KB.	L2	CO4	8
b.	(i) Explain in step by step, how would the system handle multiple interrupts when the processor is executing a particular Interrupt Service Routine (ISR)? Assume nested interrupts are enabled. (ii) Consider a computer system with a 16-bit address bus and a 256-entry interrupt vector table. Each entry in the interrupt vector table holds the address of an interrupt service routine (ISR), and the entries are aligned such that each entry occupies 4 bytes. What is the size (in bytes) of the interrupt vector table?	L3	CO3	7
2.a	Assume 4-bit signed binary numbers. (i) Add -6 to +4 and (ii) Subtract -5 from -7 using 2's complement form of binary numbers..	L3	CO3	7
b	Use Booth's Algorithm to multiply the following two 4-bit signed numbers: A = -3 and B = 6 (i) Represent the numbers in 4-bit 2's complement form. (ii) Step through the Booth's algorithm process and show the intermediate results.	L3	CO4	8
3.a	Explain in detail the working of bus arbitration using the Daisy Chain method. Provide a neat diagram to support your explanation.	L2	CO3	8
b	Discuss the basic organization of micro programmed control unit with neat circuit diagram.	L2	CO5	7

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create